Running into error 'Composition is not defined over Operators of different

https://github.com/qiskit-community/qiskit-aqua/issues/998

ROW 48

Running into error 'Composition is not defined over Operators of different dimensions, 4 and 8' #998

Closed #1048

plussoyeur opened on May 18, 2020 · edited by plussoyeur

Edits

Information

- **Qiskit Aqua version:** 0.7.1
- **Python version:** 3.7.3
- **Operating system:** NAME= elementary OS / 5.1.4 Hera

What is the current behavior?

I am trying to run a QAOA algorithm and I run into a dimensionality issue when I try to append an initial circuit. If I remove this circuit, everything works perfectly fine. I did check that my initial circuit has 4 qubits and the same holds for the operator Hc.

Steps to reproduce the problem

```
import itertools
import numpy as np
import itertools
from functools import reduce, partial

from qiskit import Aer, QuantumRegister, ClassicalRegister, execute, QuantumCircuit
from qiskit.tools.visualization import circuit_drawer
from qiskit.quantum_info import Pauli
from qiskit.aqua import QuantumInstance
from qiskit.aqua.operators import WeightedPauliOperator
from qiskit.aqua.components.initial_states import Custom
from qiskit.aqua.components.optimizers import COBYLA
from qiskit.aqua.algorithms import QAOA

np.set_printoptions(precision=3, suppress=True)


n_qubits = 2
n_system = n_qubits * 2
T = 1000

weights = np.array([[0,1],[0,0]])
p = 2

def product_pauli_z(q1, q2, coeff):
    eye = np.eye((n_system))
    return WeightedPauliOperator([[coeff, Pauli(eye[q1], np.zeros(n_system)) * Pauli(eye[q2], np.zeros(n_syste

def ising_hamiltonian(weights):
    H = reduce(lambda x,y:x+y,
            [product_pauli_z(i,j, -weights[i,j])
             for (i,j) in itertools.product(range(n_qubits), range(n_qubits))])
    #H.to_matrix()
    return H

Hc = ising_hamiltonian(weights)
print(Hc.print_details())


qr = QuantumRegister(n_system)
cr = ClassicalRegister(n_system)
backend = Aer.get_backend('qasm_simulator')


def prepare_init_state(T):
    init_circuit = QuantumCircuit(qr)
    α = 2 * np.arctan(np.exp(- 1/T))
    for i in range(n_qubits):
        init_circuit.rx(α, qr[n_qubits+i])
        init_circuit.cx(qr[n_qubits+i], qr[i])
    init_state = Custom(n_system, circuit=init_circuit)
    return init_state

initial_state = prepare_init_state(T)
print(prepare_init_state(T).construct_circuit())

optimizer = COBYLA()
qaoa = QAOA(Hc, optimizer, 2, initial_state)
```

Assignees

No one assigned

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

CircuitOp uses QuantumCircuit.compose, not .combine
qiskit-community/qiskit-aqua

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Participants

```
initial_state = prepare_init_state(T)
print(prepare_init_state(T).construct_circuit())

optimizer = COBYLA()
qaoa = QAOA(Hc, optimizer, 2, initial_state)

backend = Aer.get_backend('statevector_simulator')
quantum_instance = QuantumInstance(backend)
result = qaoa.run(quantum_instance)
```

Cryoris on May 18, 2020 · edited by Cryoris Edits Contributor ···

This seems to be a bug caused by naming explicitly creating the qubit register of the initial state, since the composing circuits with differently named registers can stack them on top of each other instead of appending them. Try running this slightly modified version where the qubit register is not explicitly defined

```
def prepare_init_state(T):
    init_circuit = QuantumCircuit(n_system)
    α = 2 * np.arctan(np.exp(- 1/T))
    for i in range(n_qubits):
        init_circuit.rx(α, n_qubits+i)
        init_circuit.cx(n_qubits+i, i)
    init_state = Custom(n_system, circuit=init_circuit)
    return init_state
```

Until we figure out where this composition error comes from.

plussoyeur on May 18, 2020 Author ···

Thanks a lot. This works. I did try that on an older version of qiskit and that did not solve the issue but this works perfectly here.

👍 1

Cryoris added type: bug on May 18, 2020

mavzolej mentioned this on Jun 11, 2020

Another "Running into error 'Composition is not defined over Operators of different dimensions'" #1034

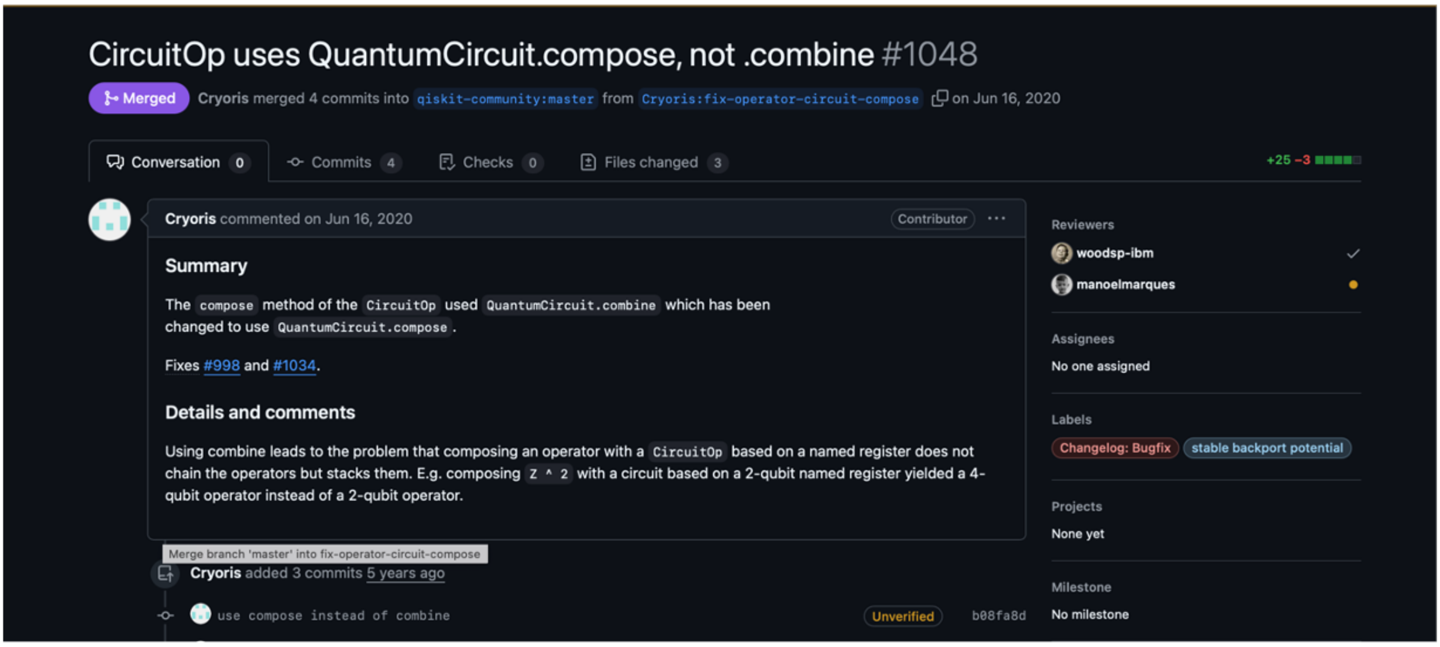
woodsp-ibm added this to To do in Aqua Core on Jun 11, 2020

Cryoris mentioned this on Jun 16, 2020

CircuitOp uses QuantumCircuit.compose, not .combine #1048

Cryoris closed this as completed in #1048 on Jun 16, 2020

Cryoris moved this from To do to Done in Aqua Core on Jun 16, 2020


CircuitOp uses QuantumCircuit.compose, not .combine #1048
Merged
Cryoris merged 4 commits into qiskit-community:master from Cryoris:fix-operator-circuit-compose on Jun 16, 2020
Conversation 0
Commits 4
Checks 0
Files changed 3
+25 −3
Cryoris commented on Jun 16, 2020
Contributor
Summary
The compose method of the CircuitOp used QuantumCircuit.combine which has been changed to use QuantumCircuit.compose .
Fixes #998 and #1034.
Details and comments
Using combine leads to the problem that composing an operator with a CircuitOp based on a named register does not chain the operators but stacks them. E.g. composing Z ^ 2 with a circuit based on a 2-qubit named register yielded a 4-qubit operator instead of a 2-qubit operator.
Merge branch 'master' into fix-operator-circuit-compose
Cryoris added 3 commits 5 years ago
use compose instead of combine
Unverified
b08fa8d
Reviewers
woodsp-ibm
manoelmarques
Assignees
No one assigned
Labels
Changelog: Bugfix
stable backport potential
Projects
None yet
Milestone
No milestone